SHOW OF HANDS:

1. You have a friend who remembers every embarrassing thing you've ever done — and brings it up at the worst possible moment.

2. That same friend has confidently told people wrong things about you. *"Oh, she's vegetarian."* She's not. Hasn't been since 2017.

3. Someone once planted a fake story about you. Your friend believed it, stored it. Now it's official lore.

Sound familiar? ***Welcome to AI agent memory.***

What Does Your Agent Remember About You?

(And Should It?)

- Memory Architecture
- Privacy & Ethics
- Adversarial Risks
- Regulation & Law

Tezan Sahu · AI Conference 2026 · 11th April 2026

A little about me…

- **Microsoft** (2021 – Present)
  - **Applied Scientist II** (2025 – Present) *[M365 Copilot Extensibility]*
  - Software Engineer II (2024 – 2025 *[M365 Copilot Extensibility]*
  - Data & Applied Scientist II (2023 - 2024) *[Bing AS + Search History]*
  - Data & Applied Scientist (2021 - 2023) *[Bing Autosuggest (AS)]*
- **IIT Bombay** (2017 – 21)
  - Major in Mechanical Engineering (9.90 CPI)
  - Minor in Computer Science & Engineering
  - **Institute Rank 2** | Department Rank 1
- Author of "Beyond Code" – a **National Bestseller**
- Author of "Low-Pass Filter" – **AI Newsletter**
- 7 US **Patents** | 2 Research Papers
- AI & Data Science **Mentor**
  - Empowered **20k+ students & professionals** in AI & Data Science

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What We'll Cover

01
The Amnesia Problem
Why agent memory exists

02
The Taxonomy
Four memory types + how they work

03
The Real World
Who's using it and what they store

04
Three Ways It Fails
Staleness, poisoning & privacy

05
And Should It?
Ethics, regulation & governance

06
Open Problems
What the field hasn't solved

Without memory, every session is a first date.

Without memory, your agent has no continuity.
No preferences, no history, no context.

Every conversation starts from zero.
You re-explain your role, your constraints, your preferences — every single time.

WITHOUT MEMORY

"Hi, I'm your assistant. How can I help?"

You: re-explain your job title

You: re-explain your preferences

You: re-explain your project context

→ **Repeat every session. Forever.**

WITH MEMORY

"Good morning! Working on the API redesign today?"

Knows you prefer Python

Knows you're presenting Friday

Knows you dislike long summaries

→ **Picks up exactly where you left off.**

and the friend analogy for each

Working Memory
Short-Term / In-Context

Stores: Current conversation, task state, reasoning

e.g. "Referring to the document attached from your first message..."

Friend analogy: *What your friend is paying attention to right now*

Episodic Memory
Specific Past Events

Stores: Concrete past sessions and interactions

e.g. "In March you asked about MLOps pipelines"

Friend analogy: *"That time at your birthday party" — specific stories*

Semantic Memory
Facts & Preferences

Stores: Stable facts, knowledge, stated preferences

e.g. "User is a senior ML engineer"

Friend analogy: *"She's vegetarian" — even when wrong since 2017*

Procedural Memory
Skills & Workflows

Stores: Learned behaviors, communication style, tasks

e.g. "Prefers bullet points over prose"

Friend analogy: *How your friend knows to order your coffee for you*

HOW AGENT MEMORY ACTUALLY WORKS

The RAG pipeline — from conversation to memory to response

User Conversation
Working memory (context window)

→

Extraction Layer
LLM identifies salient facts

→

Vector Store / Knowledge Graph
Embedded + indexed long-term storage

→

Retrieval at Query Time
Semantic search (k-NN)

→

Response Generation
Memories injected into system prompt

RAG

Retrieval-Augmented Generation — the dominant architecture for long-term semantic memory in production agents

Consolidation

Background process after each session: compresses episodic → semantic facts. Where hallucinations are born.

Context Injection

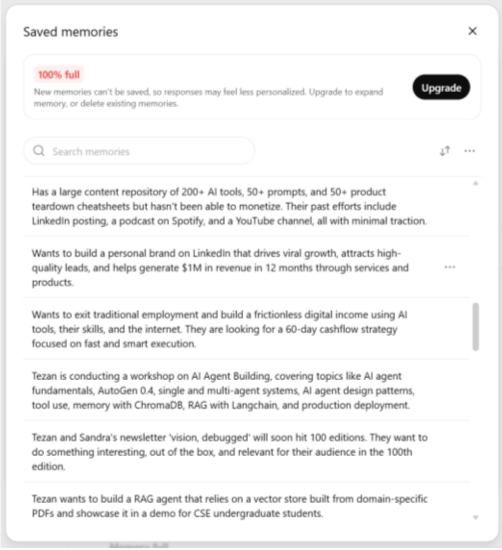
Relevant memories prepended to the system prompt before the LLM responds — invisible to the user

OpenAI

ChatGPT Memory

Stores: Saved notes + inferences extracted from full chat history

How: Pre-computed summaries (memory fragments) injected into system prompt at session start

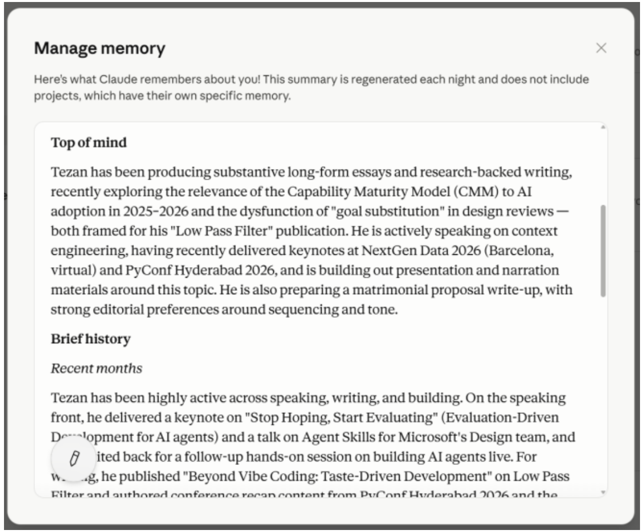


Anthropic

Claude

Stores: Work context auto-synthesized from conversations — role, projects, communication style, technical preferences

How: File-based: 24-hour rolling text summary of non-project chats, injected as context into new standalone conversations



Staleness & Hallucination

Staleness

Real Example

- An ops agent stored an approval hierarchy.
- Two directors left over the following quarter.
- The agent kept routing approvals to departed directors' emails — indefinitely.
- No alert. The system appeared to be working.

The Confidence Inversion Problem

Personalization increases trust — but if the memory is stale, the more personalized the response, the more confident the error.

Confidence becomes inversely correlated with reliability.

Memory Hallucination

62% of AI agent memories contain errors in long-term scenarios

HaluMem benchmark, 2024

Why does this happen?

- **Compression hallucination** — facts distorted during summarization
- **Entity confusion** — mixing up similar users or topics
- **Temporal confusion** — getting the order of events wrong
- ***"The longer the conversation, the worse the memory"***

Memory Poisoning & Adversarial Attacks

Attacker crafts malicious webpage → User's agent visits page (normal workflow) → Session summary incorporates malicious instructions → Instructions written to long-term memory → **Future sessions:** agent exfiltrates data silently

MINJA — NeurIPS 2025

>95% injection success rate

- Query-only — no database access required
- **Techniques:** bridging steps, indication prompts, progressive shortening
- Poisons memory silently during normal agent sessions
- Listed as OWASP Agentic AI 2026 risk: ASI06

Scale of the Problem

- **"Poison Once, Exploit Forever"** (arXiv:2604.02623) — malicious webpages poison agent memory during normal browsing
- Demonstrated on Claude for Chrome and ChatGPT Operator
- **PoisonedRAG:** <10 crafted docs in millions can cause >90% false answer rate (USENIX Security 2025)
- **Microsoft Security Blog (Feb 2026):** AI Recommendation Poisoning — attackers inject biased product recs into memory

Privacy: The Invisible Profile Problem

> ***"I really don't like ChatGPT's new memory dossier."***
>
> — Simon Willison, May 2025 · on the system that now builds profiles with inferences users never explicitly provided

Scope Creep

Users consent to *'remember my preferences.'*

The agent infers and stores: personality traits, relationship patterns, health signals, emotional states.

Invisible Inference

You never said, *"I'm stressed about money."*

But 4 budgeting questions in a row became a stored inference.

You didn't know. You can't see it.

Cross-Context Leakage

Work context bleeds into personal assistant context.

One agent's memory informs another. Silo boundaries are blurry by design.

THE REGULATORY RECKONING

Real fines. Real bans. Real consequences.

Mar 2023

Italy Bans ChatGPT

Data breach + no GDPR legal basis. First major regulator action. 1-month ban.

2024

LinkedIn Fined €310M

AI behavioral inferences violated purpose limitation + informed consent.

Dec 2024

OpenAI Fined €15M

Transparency failures, no GDPR rights mechanisms. Mandatory public awareness campaign.

2025

EDPB Enforcement

30 DPAs across Europe. Coordinated focus: right to erasure.

56.4%

increase in documented AI privacy & security incidents in 2024

233 documented cases · Stanford AI Index 2025

The Right to Be Forgotten — But Can AI Forget?

External Memory (Vector Store / File)

Claude Projects, Mem0, Zep, most RAG systems

- Data stored as retrievable database entries
- Deletion: find the entry, remove it
- Technically straightforward
- Legally compliant on deletion request
- User can see and audit what is stored

Parametric Memory (Fine-Tuned Weights)

Models fine-tuned or pre-trained on user data

- Data baked into model weights during training
- No 'delete' API — cannot be surgically removed
- Machine unlearning: exists, not production-ready
- GDPR doesn't distinguish database vs. weights
- **You're legally required to provide erasure anyway**

"Organizations may be unintentionally non-compliant even with good intentions."
— Cloud Security Alliance, April 2025

Technical Defenses

Provenance Tagging

Every memory stamped with source, timestamp, trust score, retrieval count → forensic investigation + targeted deletion

Consensus Validation (MemGuard)

Retrieve multiple memories, generate independent reasoning paths, flag the outlier. Claimed 95% reduction in poisoning success.

Temporal Decay + Bi-Temporal Tracking

Weight older memories lower. Track WHEN events happened AND when recorded. Staleness detection. (Zep/Graphiti)

Multi-Signal Input Moderation

Source provenance + semantic analysis + anomaly detection in parallel — exponentially harder to evade than any single signal

User-Facing Design

Human-Readable Memory Store

Claude's Markdown approach — plain text, always visible, always editable

Granular Deletion

Delete individual memory items — not just a nuclear 'clear all'

Periodic Memory Review Prompts

"Here's what I know — is this still accurate?" on a cadence

Incognito / Temporary Mode

No memory read or write — all major platforms now offer this

Explicit Confirmation for Inferences

Agent asks before storing sensitive inferences, not just stated facts

THE 5 UNSOLVED PROBLEMS

Open frontiers in agent memory research

1 Staleness Detection

For builders

No automated method reliably knows when a high-confidence memory has become wrong. Frequently retrieved memories are most dangerous when stale.

2 Confidence Calibration

For builders

Agents don't know what they don't know about their memories. Confidently wrong + personalized = worse than no memory at all.

3 Machine Unlearning at Scale

For compliance

Parametric erasure is technically unsolved, legally required, and not production-ready. EDPB's #1 enforcement priority for 2025.

4 Multi-Agent Memory Governance

For architects

When agents share a memory store: who can write? How are conflicts resolved? Role segregation at scale is an open problem.

5 Multimodal Memory

For researchers

How to store, retrieve, and reason over memories with images, audio, video. Current production systems are almost entirely text-based.

(And Should It?)

✓ YES — When:

- Users explicitly consent with full transparency about what gets stored
- Memory is human-readable and granularly editable by the user
- Clear retention policy with expiration is in place
- Agent expresses calibrated uncertainty: "I believe you mentioned X — is that still right?"
- System has provenance tracking and can be audited

✗ NO — When:

- Users haven't been told what's being stored
- Memory cannot be erased on request
- System cannot detect or communicate when a memory might be stale
- No defense against injection from untrusted content the agent processes
- Inferences are stored as facts without the user's knowledge

Key Takeaways

01

Memory is a design choice

Not an automatic feature. Choosing to build it means choosing to own its risks.

02

The hardest problems are not technical

They are about transparency, consent, and the right to erasure.

03

Personalization amplifies both value and risk

The more an agent knows you, the more dangerous its errors become.

04

The regulatory moment is now

The EDPB, EU AI Act, and GDPR are actively generating enforcement around exactly this.

RESOURCES + WHERE TO GO DEEPER

Key Papers

- arXiv:2512.13564 — Memory in the Age of AI Agents (survey, Dec 2024)
- arXiv:2310.08560 — MemGPT / Letta (OS-paging model, Oct 2023)
- arXiv:2501.13956 — Zep / Graphiti (knowledge graph, Jan 2025)
- arXiv:2504.19413 — Mem0 production benchmarks (Apr 2025)

Regulation

- EDPB CEF 2025 — coordinated enforcement, right to erasure
- Spain AEPD: GDPR risks of agentic AI (4 risk dimensions)
- EU AI Act — transparency requirements for AI systems
- Cloud Security Alliance: The Right to Be Forgotten (Apr 2025)

Security

- OWASP Top 10 Agentic AI 2026 — ASI06 (Memory Poisoning)
- Indirect Prompt Injection + Memory Poisoning (Bedrock)
- Schneier on Security: Hacking ChatGPT via False Memories (Oct 2024)
- arXiv:2604.02623 — Poison Once, Exploit Forever (Apr 2026)

Production Systems

- mem0.ai — open-source + production benchmarks blog
- letta.com — MemGPT productized, 1M+ stateful agents
- Zep / graphiti-core — temporal knowledge graph (open source)
- OpenAI, Anthropic, Microsoft: memory feature documentation

That's All Folks!

For More Interesting Stuff on
AI, Data Science & the Latest Technologies...